

Autodesk® Scaleform®

Scaleform 4.2

/

(Scaleform 4.2 Mobile Game Kit)

,

Scaleform

Nate Mitchell

1.01

2012 6 5

Autodesk® Scaleform® 4.2

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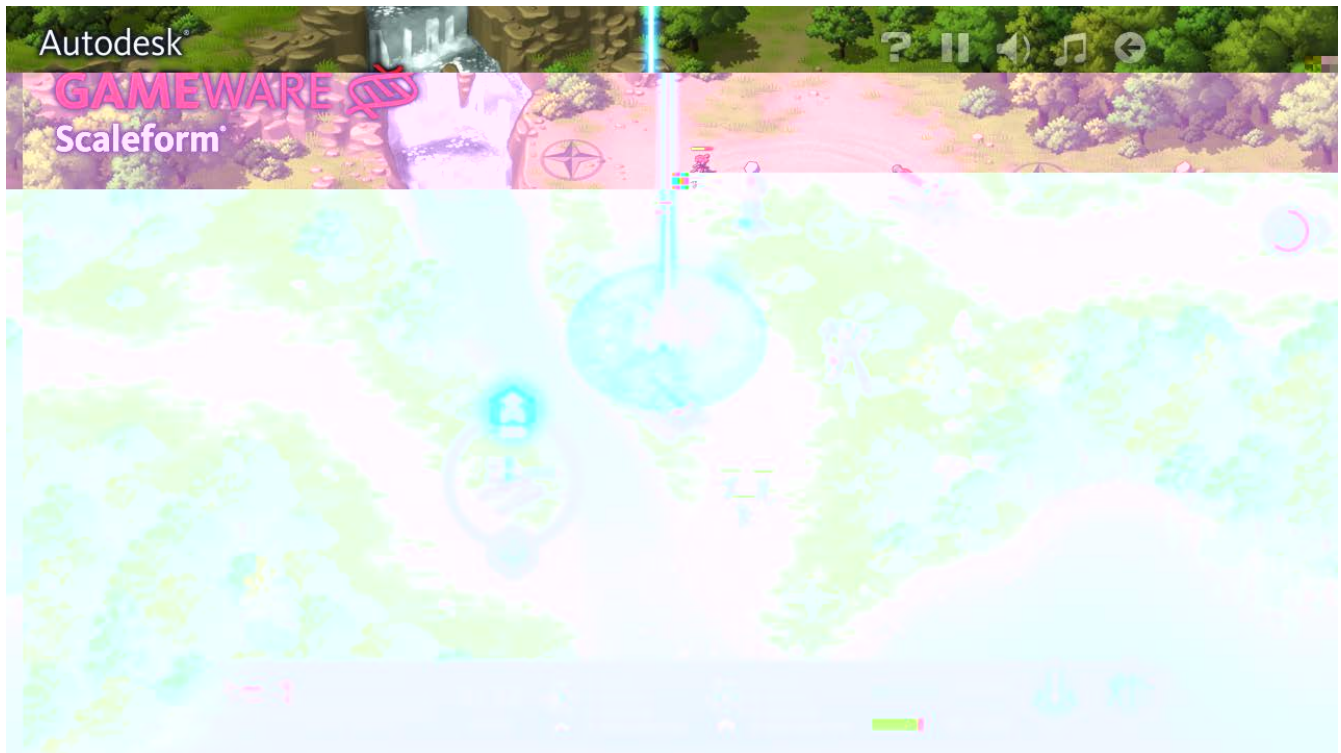
Document	Mobile Game Kit Overview
Address	Autodesk Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

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1

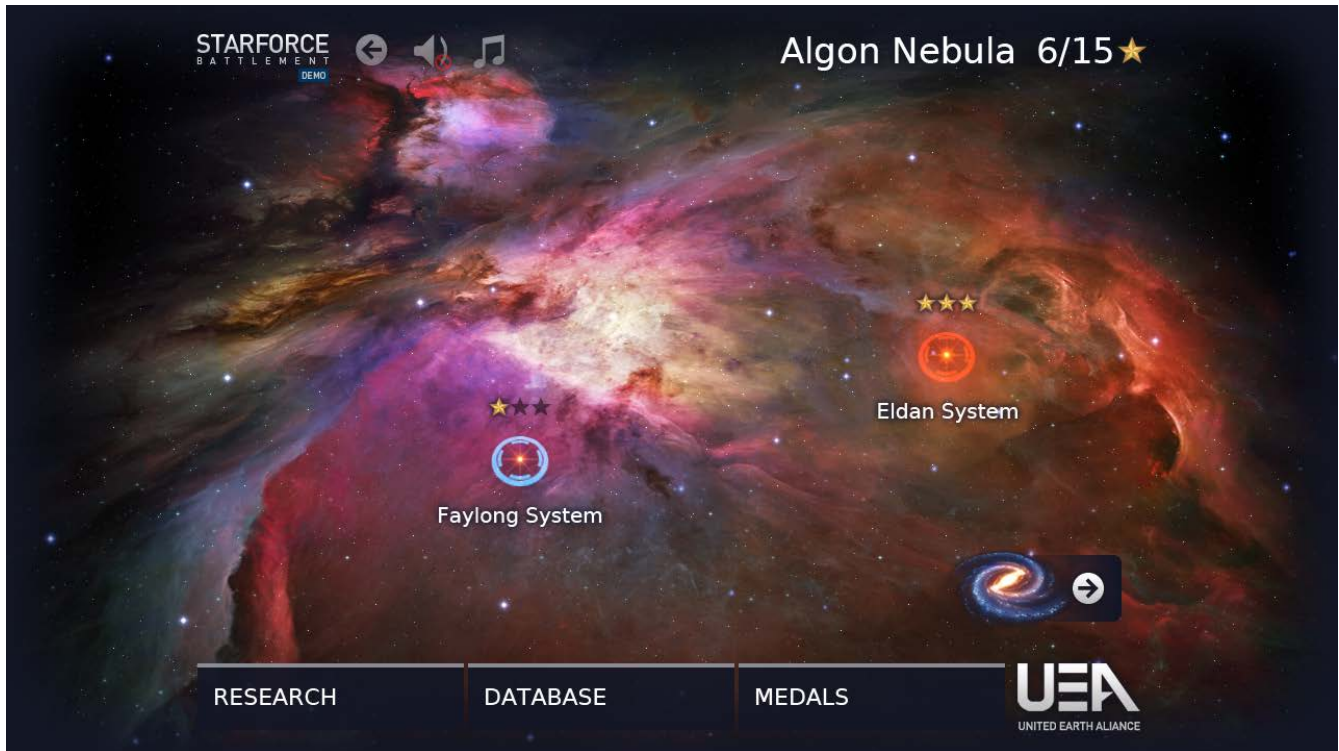
Scaleform
Autodesk® Scaleform® Starforce Battlement
ActionScript 3 , Scaleform (Shippable Mobile Player)
(PC Mac Linux iOS Android)
ActionScript 3 ,
iOS (iOS Game Center)
Adobe® Flash® (Asset) , /
Scaleform
Scaleform , Adobe Flash Player
Web , Scaleform
/ Starforce Battlement
1. Flash Professional UI
2. ActionScript 3 (FlashDevelop 4
3. Scaleform Exported .SWF
4. Scaleform GPU ,

1.1 Starforce Battlement



1 Starforce – 1

/ (/)



2 Starforce –

,Starforce Battlement

/ ,

2012 5 , iOS App Store Starforce Battlement
 iOS , Scaleform iOS

1.2

Starforce Battlement

Scaleform

- 1.
2. GPU
3. ActionScript 3
- 4.
- 5.
- 6.
7. iOS
8. (Orientation Lock) / (Orientation Handling)

 ActionScript 3

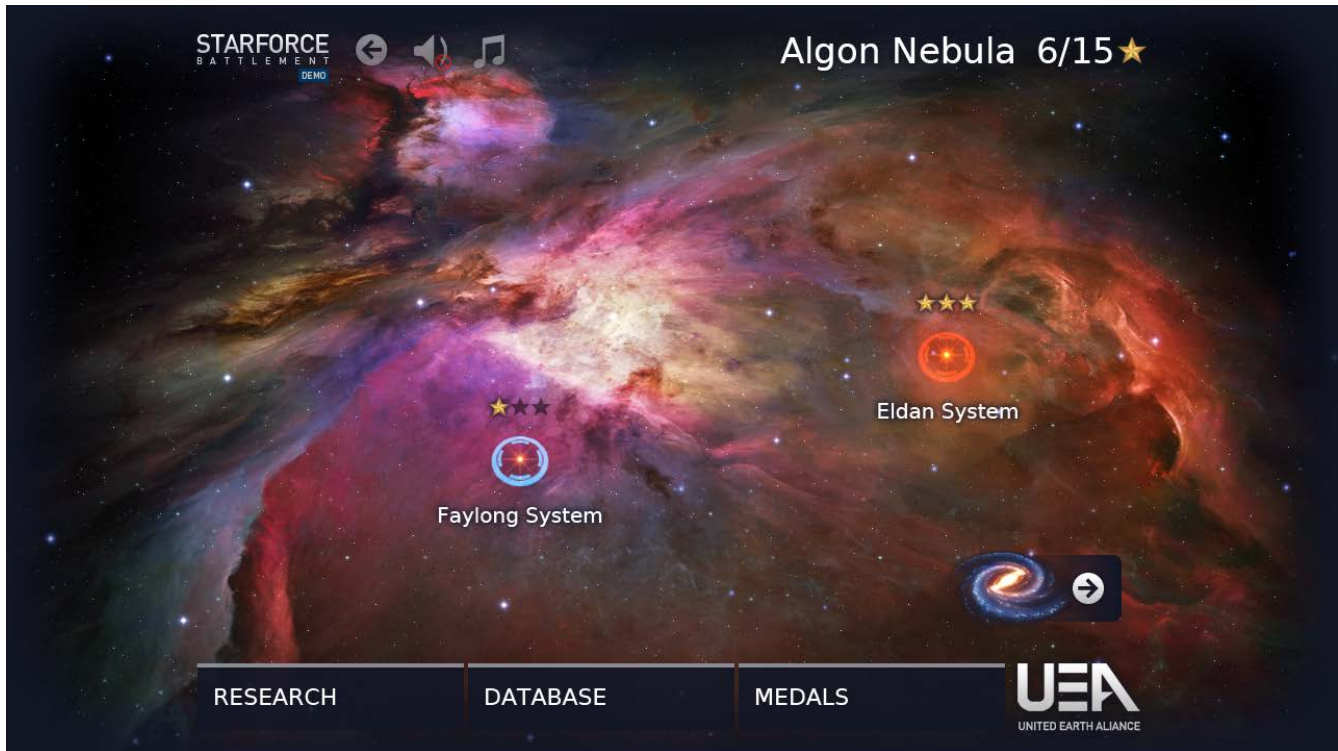
- 1.
- 2.
3. iOS
- 4.

2.1

- iOS , Xcode , iTunes SWF ()

2.2

2.2.1



3 Starforce –

```
StarforceTD.as           StarforceTD fla    Document Class(      ),
MainMainState MainMainState StarforceMenu.swf StarforceMenu.swf
,                          ActionScript 3
```

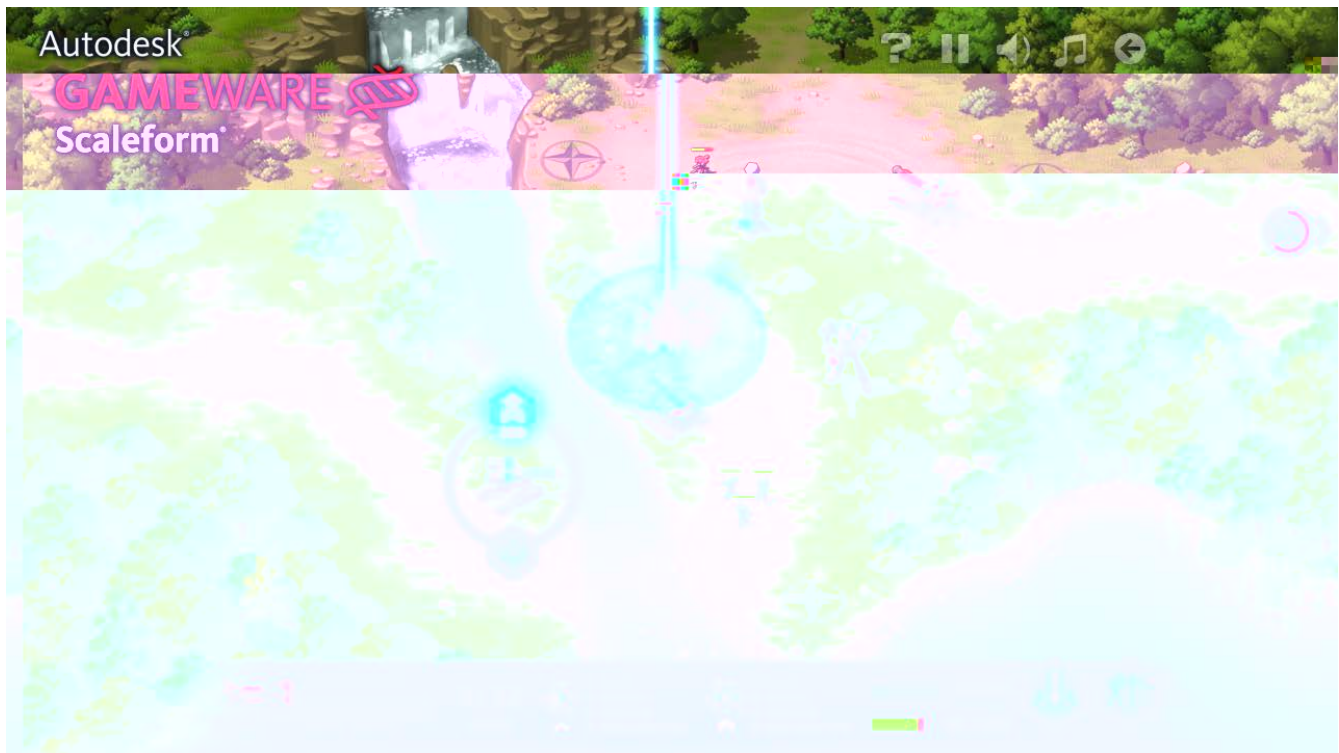
, / (galaxy/nebula)
2D ,



4 Starforce –

, (MissionDataView.as) (Mission
Data View) , (Enter Mission)
View) , X (Mission Select

HUD



5 Starforce -

Starforce Battlement

StarforceTD.swf
LevelGameState

StarforceTD.swf

,

/ ,

-- Rush Wave ()-- ,

HUD

1 2 3 ,

$$\left(\begin{array}{c} \text{ } \\ \text{ } \end{array} \right),$$

HUD (Health)

HUD

HUD : (Orbital Strike) (Deploy Mercenaries),
 ,
 (,
 ,
 ,

HUD / /

3

3.1 Flash

Starforce Battlement Flash .FLA , UI (Bitmap) , 3D (UI) FLA ActionScript

- Starforce Battlement FLA
- 1. **StarforceTD fla** – .FLA Starforce Battlement UI (Main Menu) / ActionScript (Loading Screen),
 - 2. **StarforceMenu fla** – (Mission Select) ActionScript StarforceTD.swf , ,
 - 3. **LoadingView.swf** – () ActionScript

3.2 ActionScript

3.2.1

Starforce Battlement UI ActionScript 3 , iOS Game Center , Starforce

3.2.1.1

StarforceTD.as , Flash Document Class , StarforceTD.swf (Document Classes) , Adobe Flash

3.2.1.2

```
        'GameState'(          ),
GameState,          StarforceTD          ,          GameState
    1. MenuGameState –
    2. LoadingGameState –
    3. LevelGameState –

StarforceTD          (          ),          StarforceMenu.swf
MainMenuView.as
```

3.2.1.3

```
        com.scaleform.std.menu          (Package)          (          ,
    )          ,          ,          ActionScript 3
        /          (          ,          )
        ,          GameEvent(          ) StarforceTD.as          ,
LoadingGameState          MenuGameState          LevelGameState,
LevelGameState          ,          ,
        ,          ,          LoadingGameState,
StarforceTD.swf          ,          LoadingGameState          ,
        HUD,
```

3.2.1.4

```
Level.as          ,          HUD          ,          Level.as
        ,          ,          ,          (          ,
        ,          )
Level          ,
        ,          (          ,          )
```



```

        Scaleform
        ,
        ,
        ,
        , Starforce
        MouseEvent TouchEvent MouseEvent , TouchEvent
        MouseEvent , TouchEvent.TOUCH_TAP
        MouseEvent.MOUSE_CLICK , TouchEvent.PRESS MouseEvent.MOUSE_DOWN

        TouchEvent MouseEvent , Event(
    )
        , ,target( )
        is ,

protected function onSetRallyClick( e:Event ):void {
    var point:Point;
    if (e is MouseEvent) {
        var me:MouseEvent = e as MouseEvent;
        point = new Point( me.stageX, me.stageY );
    }
    else {
        var te:TouchEvent = e as TouchEvent;
        point = new Point( te.stageX, te.stageY );
    }
}

```

3.2.2 ActionScript

```

/ , ActionScript ActionScript' (Package)

com.scaleform.std

1. abilities - ( )
2. controls - UI
3. core - (State Flow) /
4. data - ,
5. entities - ,
6. events - UI
7. fx - ( )
8. hud- HUD
9. levels - ,Level( )

```

10. loading	–	(Loading Screen)	
11. menu	–	,	UI
12. system	–		ActionScript
13. utils	–	(	)

3.2.3

3.2.3.1 /

```

( com.scaleform.std.system.SaveSystem.as )

,

ActionScript      SharedObject      SharedObject      ActionScript 3 ,
,      ActionScript      Scaleform      SharedObject
(Primitive Type)      (Array)

SharedObject
,SharedObject      sb_player_data

// Local reference to the galaxy progress. Populated from the PlayerData.
public static var galaxyProgress:Array = null;
// Constant String for the property of the SharedObject we'll modify.
protected static const GALAXY_PROGRESS_SO_PROP:String = "galaxyProgress";
// Reference to the SharedObject once we retrieve it.
protected static var _playerDataSO:SharedObject = null;

public static function load( pd:PlayerData ):void {
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ];
    pd.setGalaxyProgress( galaxyProgress );
}

public static function save( pd:PlayerData ):void {
    // Write PlayerData into SharedObject.
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = pd.getGalaxyProgress();
    _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ] = galaxyProgress;
    _playerDataSO.flush();
}

```

```

        , SharedObject.SharedObject
        SharedObject
    },

```

```

    SharedObject
    ,

```

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/net/SharedObject.html

3.2.3.2

```

        (com.scaleform.std.system.SoundSystem
        SoundEvent (
        .mp3)
    SoundEvent stage

```

```

dispatchEvent( new SoundEvent( SoundEvent.PLAY_SOUND, true, true,
"infantry_attack.mp3", 1, false ) );

```

```

        ,
        , SoundEvent
    ,

```

```

SoundEvent

```

```

dispatchEvent( new SoundEvent( SoundEvent.MUTE_SOUNDFX, true, true );

```